EXERCISE 2: AGGREGATE FUNCTIONS & GROUPING

1. COUNT () Function

```
SELECT COUNT (DISTINCT id) AS number of employees
FROM employees;
```

number of employees
10

2. SUM () Function

```
SELECT SUM (salary) AS total_salary
FROM employees
WHERE department = 'IT';
```

total_salary
220 000

3. AVG () Function

```
SELECT AVG (salary) AS avg_salary
FROM employees
WHERE department = 'HR';
```

avg_salary
49 500

4. MIN () and MAX () Functions

```
SELECT MIN (salary) AS lowest_salary
       MAX (salary) AS highest_salary
FROM employees;
```

lowest_salary	highest_salary
48000	62 000

5. GROUP BY STATEMENT

```
SELECT department,
        Sum(salary) AS total_salary
FROM    employees
GROUP BY department;
```

department	total_salary
IT	220 000
HR	99 000
Finance	119 000
Marketing	105 000

6. GROUP BY and COUNT ()

```
SELECT city,
        COUNT(*) AS number_of_employees
FROM    employees
GROUP BY city;
```

city	number_of_employees
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7. GROUP BY and ORDER BY

```
SELECT department,
        AVG(salary) AS avg_salary
FROM    employees
GROUP BY department
ORDER BY avg_salary DESC;
```

department	avg-salary
Finance	59 500
IT	55000
Marketing	52 500
HR	49 500

8. HAVING Clause

```
SELECT departments,
        sum(salary) AS total_salary
FROM employees
GROUP BY department
HAVING ~~total_salary~~ sum(salary) > 100 000;
```

department	total_salary
IT	220 000
Finance	119 000
Marketing	105000

9. Combining GROUP BY, HAVING, and ORDER BY

```
SELECT city,
        COUNT(*) AS number_of_employees
FROM employees
GROUP BY city
HAVING COUNT(*) > 1
ORDER BY number_of_employees DESC;
```

city	number_of_employees
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10. Combining Aggregate Functions

```
SELECT department,
        AVG (salary) AS avg_salary
FROM employees
GROUP BY department
ORDER BY avg-salary DESC
LIMIT 1;
```

department	avg-salary
Finance	59 500